



Recent trends in integrated bioprocesses: aiding and expanding microbial biofuel/biochemical production

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Microbial biosynthesis of fuels and chemicals represents a promising route for their renewable production. Product toxicity, however, represents a common challenge limiting the efficacy of this approach. Integrated bioprocesses incorporating *in situ* product separation are poised to help address this intrinsic problem, but suffer their own unique shortcomings. To improve and expand the utility of this versatile bioprocessing strategy, recent innovations have focused on developing more effective separation materials and novel process configurations, as well as adapting designs to accommodate semi-continuous modes of operation. As a result, integrated bioprocesses are finding new applications to aid the biosynthesis of an ever-growing list of bioproducts. Emerging applications, meanwhile, are exploring the further expansion of such designs to interface microbial and chemical catalysts, leading to new and versatile routes for the one-pot synthesis of an even greater diversity of renewable products.

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Introduction

Microorganisms serve as programmable biocatalysts for converting renewable substrates into sustainable fuel and chemical products [1,2]. Compared to traditional chemical catalysts, microbial biocatalysts offer several inherent advantages including: the flexibility to assimilate and convert complex and varied substrates (e.g. woody biomass); high selectivity, which improves efficiency while minimizing waste byproduct generation; and the ability to perform cascade reactions, facilitating the total biosynthesis of target products or building-blocks. These features, combined with ambient temperature/pressure operation,

supports the development more sustainable production processes. However, microbial biocatalysts are often less robust than chemical catalysts; an inherent shortcoming that limits their industrial potential. One common challenge arises due to product toxicity at relatively low concentrations, which effectively caps achievable performance, leading to economically unviable production metrics (i.e. titers, yields, productivities). Moreover, it also limits the potential of semi-continuous operating strategies (e.g. fed-batch), meaning greater process downtime and thus further diluted overall performance.

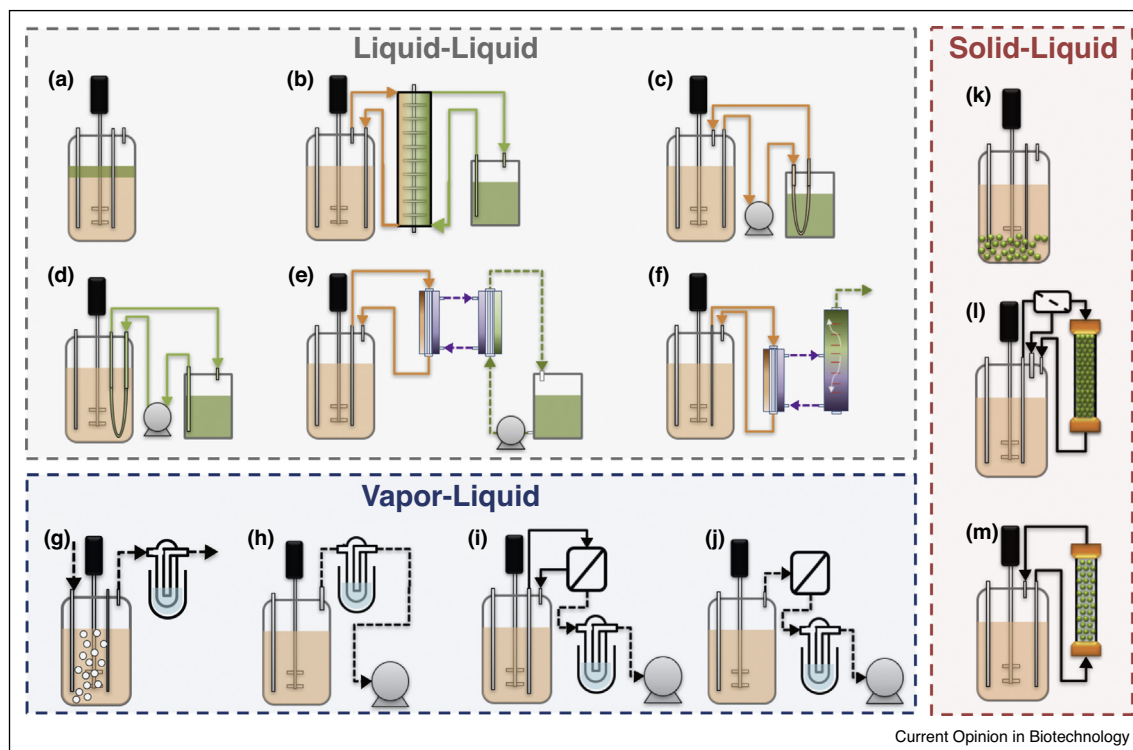
Although potential biological solutions to product toxicity continue to be actively explored (e.g. engineering strain robustness [3–6] or evolving/recruiting more tolerant chassis strains [7]; as previously reviewed in Ref. [8]), physicochemical-based strategies provide an alternative and perhaps even complementary approach. Numerous examples of ‘integrated bioprocesses’ incorporating *in situ* recovery of inhibitory bioproducts have been developed to facilitate their separation from cultures as they are produced, thereby preserving cell fitness and productivity. Since the early 1980s when initial reports of ‘extractive fermentation’ methods first surfaced in the literature, a myriad of separation mechanisms (e.g. solvent extraction, perstraction, adsorption, gas or vacuum stripping, pervaporation, electrodialysis, crystallization; as previously reviewed in Refs. [9,10]) and materials have been explored to address a range of unique bioproduction challenges (Figure 1).

As highlighted in this short review, ongoing innovations with respect to developing both advanced separation materials and novel process configurations continue to spur the maturation of integrated bioprocess technologies, facilitating their application toward the improved production of a diverse and expanding pool of useful bioproducts. Emerging applications, meanwhile, are exploring how further process integration can be leveraged to enable *de novo* synthesis of products not yet attainable by biology alone.

Novel bioproducts: new challenges for integrated bioprocesses

With ongoing innovations supporting the further maturation of metabolic engineering and systems/synthetic biology practices [11,12], streamlined biocatalyst design/construction continues to usher in a host of new, advanced biofuel and biochemical products [13,14]. Yet, given the heterologous nature of their biosynthesis, for many of these new compounds (for which native

Figure 1



Integrated bioprocess configurations developed and utilized for the *in situ* recovery of various bioproducts based on separation mechanism exploiting liquid-liquid, vapor-liquid, or solid-liquid equilibrium phenomena. (a) Direct solvent addition, (b) circulation through an external extractor, (c) perstraction with culture circulation, (d) perstraction with solvent circulation, (e) in-line hollow fiber membrane perstraction, (f) reactive extraction using a hollow fiber membrane-coupled with distillation, (g) gas stripping, (h) vacuum stripping, (i) pervaporation, (j) vapor permeation, (k) direct adsorbent addition, (l) packed bed adsorption, (m) expanded bed adsorption.

tolerance mechanisms have yet to be evolved), achievable production metrics are commonly limited by end-product toxicity. In such cases, one potentially useful strategy for improving their production involves reconfiguration of established integrated bioprocess designs. This is facilitated by the fact that, although perhaps unique in chemical functionality and/or structure, many novel bioproducts still share the same exploitable bulk physiochemical attributes (e.g. volatility, hydrophobicity) of various, preceding natural fermentation products (e.g. ethanol, *n*-butanol). Accordingly, this basic approach has been adopted to improve production of a range of emerging bioproducts, including higher alcohols (e.g. isobutanol, gas stripping [15]; 3-methyl-1-butanol, extraction [16]) and aldehydes (e.g. isobutyraldehyde, gas stripping [17]), volatile esters (e.g. ethyl acetate, gas stripping [18]; butyl butyrate [19], extraction), organic acids (e.g. itaconic acid, reactive extraction [20*]; lactic acid, ultrafiltration and nanofiltration followed by ion exchange [21]), long-chain free fatty acids (FFAs) (adsorption [22]), and aromatics (styrene, gas stripping or extraction [23,24]; 2-phenylethanol, extraction [25]). For example, several studies have demonstrated how

bacterial phenol production can be enhanced via the use of *in situ* solvent extraction. In one particularly successful application, phenol production by engineered *Escherichia coli* was improved nearly 10-fold versus traditional batch culture via integrated extraction with glycerol tributyrates and under fed-batch condition [26]. Another class of compounds, for which the use of integrated bioprocess designs has proven particularly useful is isoprenoids. Solvent extraction, for example, has been successfully applied to improve production of amorph-4,11-diene [27] and 3-methyl-3-buten-1-ol [28] by *E. coli*, limonene and α -bisabolene by *Synechococcus* sp. PCC 7002 [29], and farnesene by *Saccharomyces cerevisiae* [30]. In one notable case, the sesquiterpene alcohol α -bisabolol was produced from glucose by engineered *E. coli* and extracted using vegetable oil as a cheap and 'green' solvent [31]. Though inhibition occurs in the presence of as little as 1 g/L α -bisabolol, as a result of its *in situ* extraction, final titers of up to 9.1 g/L were achieved during fed batch operation, along with >98% overall recovery. Lastly, with a boiling point close to that of growth temperature of *E. coli*, isoprene readily evolves from cultures, making gas stripping an effective mechanism for its *in situ*

recovery [32]. This approach has been reported to support the production of >60 g/L isoprene under fed-batch conditions at the industrial scale [33].

Advanced separation materials and novel process configurations: new tools and strategies for *in situ* bioproduct recovery

Although integrated bioprocesses offer potential solutions to overcoming the performance-limiting effects of product toxicity, depending upon the separation mechanism and materials employed, they too can impose their own inherent incompatibilities and shortcomings. To address such limitations, recent efforts in the field have accordingly sought to develop both advanced separation materials and novel process configurations. For instance, with tunable chemistries and extremely low vapor pressures [34], ionic liquids (ILs) have been proposed as effective alternatives to traditional organic solvents for a variety of *in situ* bioproduct separations [35]. In general, however, high cytotoxicity of many ILs remain as a critical factor limiting their wider use [36]. To address this, in addition to developing more IL-tolerant strains [37–39], increasingly biocompatible IL solvents are also being developed. Bacon *et al.*, for example, recently developed and characterized a series of water-insoluble, solid imidazolium-based polyionic liquids (PILs) as butanol absorbents that displayed high biocompatibility with *S. cerevisiae* and competitive separation performance [40^{*}]. Alternatively, by providing a barrier between the culture and solvent, membrane extraction (i.e. perstraction) removes this biocompatibility constraint altogether, allowing greater flexibility in extraction conditions. This approach has proven particularly useful in the case of reactive extraction, a promising mechanism for the *in situ* recovery of value-added organic acids. Ge *et al.*, for example, have applied reactive extraction to upgrade ethanol to *n*-caproate using a mixed microbial culture [41^{**}]. In this case, a two-module, in-line hollow fiber membrane perstraction system was developed to simultaneously perform both forward extraction and backward extraction of produced *n*-caproate. Remarkably, this strategy allowed the process to operate continuously for a period of 55 days, accumulating *n*-caproate at a total level nearly 10-times its inhibitory limit. Subsequent integration of an additional membrane electrolysis cell, meanwhile, enabled direct recovery of free *n*-caproic acid in an integrated bioprocess operated continuously for 83 days [42]. Lastly, Saboe *et al.* recently reported an alternative process for the production and *in situ* recovery of a mixture of short-chain to medium-chain carboxylic acids that incorporated 1) reactive extraction using hollow fiber membrane and 2) distillation to release neat free acids and regenerate the solvent for reuse [43^{*}].

While membrane-based strategies typically boast greater biocompatibility than liquid solvents, they in turn are often prone to biofilm formation or other forms of fouling.

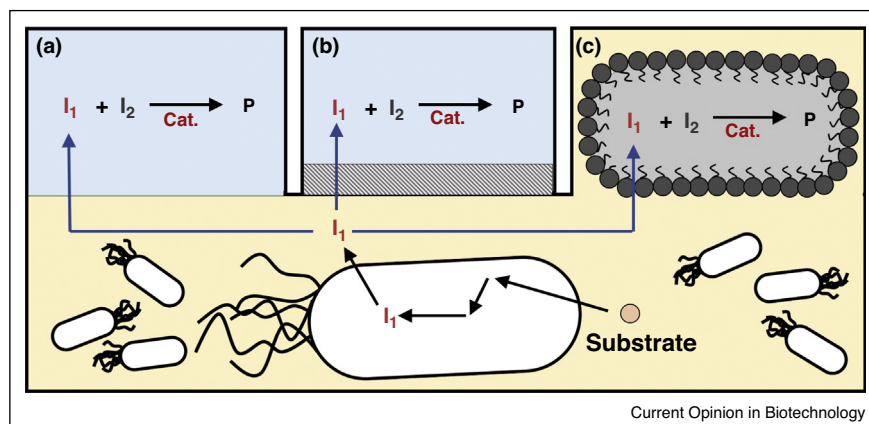
Solutions to this problem are again likewise being addressed through both biocatalyst and separation materials development. Engineering and use of self-flocculating yeast strains, for example, prevent their interaction with and fouling of membrane surfaces during pervaporation [44], whereas surface modifiers have also been used to reduce various forms of fouling [45,46]. Materials development efforts have furthermore focused heavily on improving separation performance (i.e. permeability, flux, separation factor). One promising approach to improve separation performance while still retaining desired mechanical and chemical stability attributes during pervaporation, for example, involves the fabrication of hybrid or mixed matrix membranes (MMMs). MMMs are synthesized by incorporating filler materials with high organophilic selectivity (e.g. zeolites [47,48], carbon nanotubes (CNTs) [44], and metal organic framework (MOF) materials [49]) into a durable polymer matrix. Tuning of separation behavior is achieved by the modification of particle chemistry and size as well as membrane synthesis conditions — strategies used to develop effective MMMs for ethanol and butanol pervaporation, for example [49].

Adsorption similarly offers high inherent biocompatibility while representing a low energy *in situ* separation mechanism amenable for use with hydrophobic or anionic (e.g. organic acids) bioproducts. Numerous different classes of adsorbents have been investigated for this purpose, with materials research focused heavily on developing adsorbents with higher capacities to reduce material costs. Capacity improvements have focused on altering the chemistry of the adsorbent matrix, density or nature of associated adsorption sites, controlling pore size/structure, and/or developing materials with extremely high-specific surface areas [50]. At the same time, novel configurations (e.g. expanded bed adsorption [51,52]) and operating modes (e.g. simulated moving bed [53]) are seeking to facilitate integration of effective adsorbents with bioreactor processes in ways that promote the prospect of semi-continuous operation. Magnetically functionalized adsorbents have also recently been developed for the same purpose, offering an alternative mechanism for their facile separation from the process with minimal disruption. For instance, a series of high surface area, mesoporous carbon powders with embedded NiO nanoparticles were recently developed and used for butanol recovery [54], whereas other magnetically functionalized adsorbents have been used to facilitate *in situ* protein purification from *Bacillus megaterium* cultures [50] and will likely find many future uses.

Interfacing microbial and chemical catalysis: a new frontier for integrated bioprocesses

In modern industrial biotechnology, additional downstream processing is routinely employed to convert microbially derived ‘building-blocks’ into biofuels [55],

Figure 2



Integrating chemical catalysis with microbial bioproduction for one-pot synthesis of a target final product (P) from a biosynthesized intermediate (I_1) and additional co-substrate (I_2 ; added or co-produced). Demonstrated or putative integrated bioprocess configurations include **(a)** direct addition of a biocompatible catalyst, **(b)** catalyst sequestration in non-aqueous, biocompatible solvent phase, and **(c)** catalyst sequestration in biocompatible micelles. Black lines indicate enzymatic or chemical reaction steps whereas blue dashed lines indicate mass transfer processes.

bioplastics [56], pharmaceuticals [57], and many other final products of interest [58]. In such cases, synthetic chemistry typically serves to ‘pick up where biology leaves off’. Although segregation of individual process steps is typically essential to ensure mutual compatibility and optimal function of both biological and chemical catalysts under their own unique and dissimilar conditions, in certain cases, the separation process employed as part of an integrated bioprocess can furthermore provide the spatial isolation necessary to simultaneously support the optimal function of both. This provides unique opportunities to synchronously and efficiently couple both biocatalytic and chemocatalytic steps within a single process or, in some cases, a single vessel.

Anbarasan *et al.* demonstrated the conversion of acetone-butanol-ethanol (ABE) mixtures from *Clostridium acetobutylicum* into C_7 – C_{15} alkanes (diesel and jet fuels precursors) at 93% yield over a Pd/C– K_3PO_4 solid catalyst at 145°C [59]. In this case, glyceryl tributyrates was used to extract the ABE products, and reused following distillation to recover the reaction products. Lennen *et al.* developed a similar, two-stage process for producing *n*-undecane from C_{12} FFAs initially synthesized from sugar by engineered *E. coli* [60].

In the above examples, spatial separation was key to preserving performance of both precursor bioproduction and subsequent chemocatalytic steps due to mutually incompatible reaction conditions. In select cases, however, where such conditions suitably align, it is possible to perform both steps within a single vessel (Figure 2). Wallace and Balskus, for example, recently reported a pair of studies focused on the one-pot synthesis of non-natural phenyl cyclopropanes directly from glucose [61]. In this case, biocompatible iron

(III) phthalocyanine catalyst was added to the aqueous medium to enable efficient olefin cyclopropanation using styrene, produced from glucose by engineered *E. coli*. A series of vitamin E-derived surfactant micelles were subsequently developed to accommodate heterogeneous, organic-soluble transition metal catalysts, reducing biocompatibility and catalyst inactivation concerns [62]. As these biocompatible micelles may be capable of accommodating other heterogeneous catalysts, future integration of additional chemistries may also be possible.

Conclusions

Integrated bioprocesses continue to provide an effective strategy toward improving the production of a diverse range of bioproducts. The development of advanced separation materials and novel configurations have both been and will remain vital toward further advancement of this bioprocessing strategy, including at both the research and industrial scales. Future applications, meanwhile, will expand the approach through further integration of additional synthetic chemistries, enabling one-pot production of previously unattainable molecules from renewable resources.

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